

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 12 HSC Assessment 2 Term 1 2023

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 75 minutes
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators may be used
- Show all necessary working
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination

Total marks –43

Section I Pages 2 – 3

8 marks

Attempt Questions 1 – 8

Answer on the Objective Response Answer Sheet provided

Section II Pages 4 – 7

35 marks

Attempt Questions 9 – 12. Start each question on a new page. Write your student number on page

Question	1-8	9	10	11	12	Total
Total	/8	/9	/9	/9	/8	/43

This assessment task constitutes 30% of the Higher School Certificate Course School Assessment

Section I

8 marks

Allow about 10 minutes for this section

Use the Multiple Choice answer sheet for Questions 1 – 8

1. How many ways 4-digit numbers can be formed using 0 to 9 without repetition?

- (A) 4536
- (B) 5040
- (C) 9000
- (D) 10000

2. Given that $y = x^2 - x^3$ and $\frac{dx}{dt} = 3$, find $\frac{dy}{dt}$.

- (A) $3x^2 - x^3$
- (B) $6x - 9x^2$
- (C) $3(x^2 - x^3)$
- (D) $2x - 3x^2$

3. Which expression is equivalent to $n! + (n-1)! + (n-2)!$?

- (A) $(n-2)!$
- (B) $(n-1)!(n^2 - 1)$
- (C) $n^2(n-2)!$
- (D) $(n-1)!(n-2)$

4. Let $A = \{1, 2, 3, 4, 5\}$. How many subsets of A contain **at least** 3 digits?

- (A) 10
- (B) 15
- (C) 16
- (D) 32

5. Which represents the general term of the expansion of $(2x + x^2)^5$?
- (A) ${}^5C_r \times 2^{5-r} \times x^r$
- (B) ${}^5C_r \times 2^r \times x^{5+r}$
- (C) ${}^rC_5 \times 2^{r-5} \times x^{3r-5}$
- (D) ${}^5C_r \times 2^{5-r} \times x^{5+r}$
6. In how many ways can 7 different charms be arranged on circular bracelet?
- (A) 720
- (B) 5040
- (C) 2520
- (D) 360
7. Given $y = 1000 + 300e^{0.2t}$, what is $\frac{dy}{dt}$?
- (A) $60e^t$
- (B) $0.2(y - 1000)$
- (C) $0.2y$
- (D) $0.2(y + 1000)$
8. How many students are needed in a school to guarantee that there are at least two students who have the same first two initials?
- (A) 676
- (B) 677
- (C) 672
- (D) 680

End of Section I

Section II

35 marks

Attempt Questions 9 - 12

Allow about 60 minutes for this section

Begin each question on a new answering booklet, indicating the question number.

Extra writing paper is available.

Question 9 (9 marks) Start a new answering booklet. marks

(a) Find the number of arrangements of SAMUEL if the vowels and consonants alternate. 1

(b) A committee of 5 is to be formed from a group of 4 women and 5 men. 1
How many committees can be formed by more men than women?

(c) Show that ${}^nC_r + {}^nC_{r-1} = \frac{(n+1)!}{r!(n-r+1)!}$ 2

(d) A pool is initially full. It's drained so that at time t minutes the volume of water, V , in kL, is given by

$$V = 50 \left(1 - \frac{t}{40} \right)^2 \quad \text{for } 0 \leq t \leq 40.$$

(i) How much water was initially in the pool? 1

(ii) Show that it takes 20 minutes for the tank to be one-quarter full. 2

(iii) At what rate was the water draining out the pool when it was one-quarter full? 2

Question 10 (9 marks)

(a)

(i) Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$, 2

where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$.

(ii) Hence, or otherwise, solve $\cos x - \sqrt{3} \sin x = 1$, for $0 \leq x \leq 2\pi$. 2

(b) A virus outbreak during a period of time followed an exponential increase model

$$N = Ae^{kt}$$

in which N is the number of people, t is the number of days, A and k are positive constants.

(i) Initially, 20 people were infected. At the end of 4 weeks, 3

12000 people were infected. Find the value of k .

Answer correct to 2 decimal places.

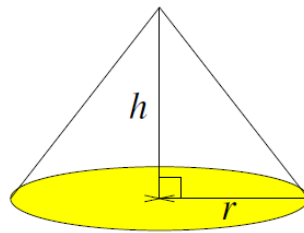
(ii) On which day will the number N of infections first exceed one million? 2

Question 11 (9 marks)

- (a) Evaluate the coefficient of x^2 in the expansion of $(5+x)(3-2x)^6$. **3**
- (b) By expressing $\sin x$ and $\cos x$ in terms of t , where $t = \tan \frac{x}{2}$, solve **3**
 $3\sin x + \cos x = -1$ for $0 \leq x \leq 2\pi$. Answer correct to 1 decimal places.
- (c) Solve the equation $\sin 5x + \sin x = \sin 3x$ for $0 \leq x \leq \pi$. **3**

Question 12 (8 marks)

- (a) Arrange A, B, C, D and E in a circle. How many ways can the circle be arranged so that C and D do not sit together. **1**
- (b) Sand pouring from a high conveyor belt is forming a cone with base diameter three times the height. Let h metres be the height of the pile, and r metres be its base radius, t minutes after pouring begins.



- (i) Show that $V = \frac{2\pi r^3}{9}$ and $\frac{dV}{dr} = \frac{2\pi r^2}{3}$. **2**
- (ii) Given the sand is being poured at a constant rate of $3 \text{ m}^3/\text{min}$, **2**
find the rate at which the radius is increasing when the radius is 6 metres.
Answer in exact value.
- (iii) At what rate must the sand be poured onto the pile if it is required that **3**
the base area increases at $12 \text{ m}^2/\text{min}$ when the base radius is 6 metres?

End of Examination

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Solutions

Solutions	Marker's Comments
Question 1 A $9 \times 9 \times 8 \times 7 = 4536$	50 out of 75 students chose answer B which did not account for the first digit not being a zero
Question 2 B $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ $= (2x - 3x^2) \times 3$ $= 6x - 9x^2$	Ax3 Cx2 Dx1
Question 3 C $(n-2)!(n(n-1) + (n-1) + 1)$ $= (n-2)!(n^2 - n + n - 1 + 1)$ $= (n-2)!n^2$	Ax2 Bx6 Dx6
Question 4 C ${}^5C_3 + {}^5C_4 + {}^5C_5 = 16$	Ax4 Bx6
Question 5 D ${}^5C_r (2x)^{5-r} (x^2)^r$ $= {}^5C_r \times 2^{5-r} x^{5-r+2r}$ $= {}^5C_r \times 2^{5-r} x^{5+r}$	Ax12 Bx2
Question 6 D $\frac{6!}{2} = \frac{720}{2} = 360$	Ax7 Bx1 Cx6
Question 7 B $\frac{dy}{dt} = 0.2 \times 300e^{0.2t}$ $= 0.2(y - 1000)$	Ax4 Cx1 Dx2
Question 8 B $26 \times 26 + 1 = 677$	Ax4 Bx1 Dx2

Solutions	Marker's Comments
Question 9 (a) Vowels: A, U, E Consonants: S, M, L The arrangement can be either VCVCVC or CVCVCV $\therefore 3! \times 3! \times 2 = 72$	There were quite a few students who didn't multiply by 2 here. Aw 1 for correct answer or correct numerical expression. (CNE)
(b) 5 men + 0 women: 5C_5 4 men + 1 women: ${}^5C_4 \times {}^4C_1$ 3 men + 2 women: ${}^5C_3 \times {}^4C_2$ $\therefore {}^5C_5 + {}^5C_4 \times {}^4C_1 + {}^5C_3 \times {}^4C_2 = 81$	There were quite a few students who didn't include 5 men and 0 women as a possibility. Aw 1 for correct answer or correct numerical expression. (CNE)
(c) ${}^nC_r + {}^nC_{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}$ $= \frac{n!(n-r+1) + n!r}{r!(n-r+1)!}$ $= \frac{n!(n-r+1+r)}{r!(n-r+1)!}$ $= \frac{n!(n+1)}{r!(n-r+1)!}$ $= \frac{(n+1)!}{r!(n-r+1)!}$	There were many who took a lot of work to get the LCD and some who struggled as they simply multiplied the factorials together w/o realising that: $r! = r \times (r-1)!$ and $(n-r+1)! = (n-r+1) \times (n-r)!$ This skill needs to be practised before your Trial and HSC exam. It is very important that you understand and can do this properly. Aw 2 for correct show procedure Aw 1 for finding LCD.
(d) (i) $t = 0, V = 50 \left(1 - \frac{0}{40} \right)^2$ $= 50 \text{ kL}$	Well done by nearly all students Aw 1 for correct answer or correct numerical expression. (CNE)
(ii) $50 \times \frac{1}{4} = 50 \left(1 - \frac{t}{40} \right)^2$ $\frac{1}{4} = \left(1 - \frac{t}{40} \right)^2$ $\pm \frac{1}{2} = 1 - \frac{t}{40}$ $\frac{t}{40} = \frac{1}{2}, \frac{3}{2}$ $t = 20, 60$ $\therefore t = 20 \quad (0 \leq t \leq 40)$	This question posed a few issues depending on method. The easy way was to substitute $t=20$ and show the volume was 12.5kL and this was $\frac{1}{4}$ of the original volume. Aw 2 for correct answer and relating to original volume. Aw 1 for 12.5kL w/o showing/relating this is $\frac{1}{4}$ of original. The other way is the more conventional method however there were a lot of students who disregarded the – without a correct reason due to the domain given. This should not be done to the final step. However, this was allowed, this time. It may not in the future. Aw 2 for this method with or w/o the +/-

Solutions	Marker's Comments
Question 9 (d) (iii) $\begin{aligned}\frac{dV}{dt} &= 50 \times 2 \left(1 - \frac{t}{40}\right) \left(-\frac{1}{40}\right) \\ &= \frac{-100}{40} \left(1 - \frac{t}{40}\right) \\ &= \frac{-5}{2} \left(1 - \frac{t}{40}\right) \\ t = 20, \quad \frac{dV}{dt} &= \frac{-5}{2} \left(1 - \frac{20}{40}\right) \\ &= -1.25 \text{ kL/min}\end{aligned}$ This means the volume of the pool is decreasing @ 1.25kL/min	This question was generally done well by most students. A few need to brush up on the skill of differentiating using the chain rule. Aw 2 for correct answer of -1.25kL/min. Bare in mind that you may need to interpret what the minus sign means properly in further assessments. i.e. This means the volume of the pool is decreasing @ 1.25kL/min, not decreasing at -1.25kL/min. Aw 2 for correct answer Aw 1 for correct working of the chain rule.
Question 10 (a) (i) $\begin{aligned}\cos x - \sqrt{3} \sin x &= R \cos(x + \alpha) \\ &= R \cos x \cos \alpha - R \sin x \sin \alpha \\ R &= \sqrt{1^2 + (\sqrt{3})^2} = 2 \\ \tan \alpha &= \sqrt{3}, \quad \alpha = \frac{\pi}{3} \\ \therefore \cos x - \sqrt{3} \sin x &= 2 \cos\left(x + \frac{\pi}{3}\right)\end{aligned}$	Done well by most students
(ii) $\begin{aligned}\cos x - \sqrt{3} \sin x &= 1 \quad (0 \leq x \leq 2\pi) \\ 2 \cos\left(x + \frac{\pi}{3}\right) &= 1 \\ \cos\left(x + \frac{\pi}{3}\right) &= \frac{1}{2} \\ x + \frac{\pi}{3} &= \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3} \quad \left(\frac{\pi}{3} \leq x + \frac{\pi}{3} \leq 2\pi + \frac{\pi}{3}\right) \\ \therefore x &= 0, \frac{4\pi}{3}, 2\pi\end{aligned}$	Many students did not consider the domain $\frac{\pi}{3} \leq x + \frac{\pi}{3} \leq 2\pi + \frac{\pi}{3}$ and therefore missed the solution $x = 2\pi$

(b) (i) $t = 0, N = 20$ $20 = Ae^{k(0)}$ $A = 20$ $t = 4 \times 7 = 28, N = 12000$ $12000 = 20e^{k(28)}$ $e^{28k} = 600$ $28k = \ln 600$ $k = \frac{\ln 600}{28}$ $= 0.22846\dots$ ≈ 0.23	There were quite a few careless numerical errors e.g. $\frac{12000}{20} \neq 60$ and some understanding errors involved with working with logs. If students are making errors with converting from exponential form to log form $e^{28k} = 600$ $28k = \ln 600$ they may have greater success by taking log of both sides instead. $e^{28k} = 600$ $\ln e^{28k} = \ln 600$ $28k \ln e = \ln 600$ $28k \times 1 = \ln 600$ $28k = \ln 600$
Solutions	
Question 10 (b) (ii) $20e^{\frac{\ln 600}{28}t} > 1000000$ $e^{\frac{\ln 600}{28}t} > 50000$ $\frac{\ln 600}{28}t > \ln 50000$ $t > \frac{28 \ln 50000}{\ln 600}$ $t > 47.3592\dots$ On the 48th day, N will exceed 1 million.	Most students dealt with the equation rather than the inequation to find $t=47.35925019$ if they used the exact value of k or $t=47.04251428$ if they used $k=0.23$ which is fine but then they did not interpret their answer correctly to state the 48th day.
Question 11 (a) $(5+x)(3-2x)^6$ $= (5+x)^6 C_r 3^{6-r} (-2)^r x^r$ $= 5 \times {}^6C_r 3^{6-r} (-2)^r x^r + x \times {}^6C_r 3^{6-r} (-2)^r x^r$ $= 5 {}^6C_r 3^{6-r} (-2)^r x^r + {}^6C_r 3^{6-r} (-2)^r x^{r+1}$ $r = 2, \quad 5 {}^6C_2 3^{6-2} (-2)^2 x^2 = 24300x^2$ $r = 1, \quad {}^6C_1 3^{6-1} (-2)^1 x^{1+1} = -2916x^2$ $24300x^2 - 2916x^2 = 21384x^2$ $\therefore$ the coefficient of x^2 is 21384.	Many students were able to write the general form correctly. Some students couldn't multiply $(5+x)$ across and add the correct coefficients. Some students expanded $(3-2x)^6$ and then multiple with $(5+x)$ to find the correct terms, which also gives the same answer. Some students are still not familiar with binomial expansion.

(b) ii) Let $t = \tan \frac{x}{2}$, $\frac{3 \times 2t}{1+t^2} + \frac{1-t^2}{1+t^2} = -1$ $6t - 1 - t^2 = -1 - t^2$ $6t = -2$ $t = -\frac{1}{3}$ $\tan \frac{x}{2} = -\frac{1}{3}, \quad 0 \leq x \leq 2\pi$ $\frac{x}{2} = \pi - \tan^{-1} \frac{1}{3}$ $x = 5.6396\dots$ $x \approx 5.6$ Check $x = \pi$, $3 \sin \pi + \cos \pi = 0 - 1 = -1$ $\therefore x \approx 5.6 \text{ or } x = \pi$	Many students managed to solve for t correctly. Some students didn't find the correct solutions for x in the domain. Some students forgot to check $x = \pi$.
(c) $\sin 5x + \sin x = \sin 3x \quad (0 \leq x \leq \pi)$ $2 \sin 3x \cos 2x = \sin 3x$ $\sin 3x(2 \cos 2x - 1) = 0$ $\sin 3x = 0, \quad \cos 2x = \frac{1}{2}$ $3x = 0, \pi, 2\pi, 3\pi, \quad 2x = \frac{\pi}{3}, \frac{5\pi}{3}$ $\therefore x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{\pi}{6}, \frac{5\pi}{6}$	Many students were not able to use appropriate angle formula to convert the equation into a form to solve for x. The product to sum formula is on the formula sheet. Some students didn't find the solutions in the correct domain for $\sin 3x$ and $\cos 2x$.
Question 12 (a) $2 \times 3 \times 2 = 12$	About 25% of students did not get the answer because they forgot to switch the two people around.
(b) (i) $2r = 3h, h = \frac{2}{3}r$ $V = \frac{1}{3}\pi r^2 h$ $= \frac{1}{3}\pi r^2 \left(\frac{2}{3}r\right)$ $= \frac{2}{9}\pi r^3$ $\frac{dV}{dr} = \frac{2}{9} \times 3\pi r^2$ $= \frac{2}{3}\pi r^2$	Some students were unable to get an equation for h in terms of r and therefore were unable to get an equation for V. This part was done well!

(ii) $\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$ $= \frac{3}{2\pi r^2} \times \frac{dV}{dt}$ sub $\frac{dV}{dt} = 3, r = 6$ $\frac{dr}{dt} = \frac{3}{2\pi(6)^2} \times 3$ $= \frac{1}{8\pi} \text{ mm/min}$	Done well!
(iii) $A = \pi r^2, \frac{dA}{dr} = 2\pi r,$ $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dA} \times \frac{dA}{dt}$ $= \frac{2}{3} \pi r^2 \times \frac{1}{2\pi r} \times \frac{dA}{dt}$ sub $\frac{dA}{dt} = 12, r = 6$ $\frac{dV}{dt} = \frac{2}{3} \pi (6)^2 \times \frac{1}{2\pi(6)} \times 12$ $= 24 \text{ mm}^3/\text{min}$	Poorly done! Many students did not get an expression for $\frac{dr}{dt}$ using the conditions for the area of the base.

End of Paper